

MATLAB/SIMULINK PROJECT REPORT

# Intelligent EV Charging and Vehicle-to-Grid Energy Management in a Smart Grid

A multi-timescale simulation framework for renewable-aware charging, peak reduction, battery state-of-charge management, and bidirectional grid support.

**24 h**

supervisory horizon

**15 min**

scheduling interval

**50 EVs**

aggregate fleet

**V2G**

bidirectional support

SMART GRID | RENEWABLE ENERGY | EV FLEET | V2G

## 01 Executive summary

This project develops a MATLAB/Simulink framework for coordinating electric-vehicle charging and vehicle-to-grid operation in a smart distribution network. The supervisory model evaluates a complete day using 15-minute intervals and makes charging decisions from feeder demand, renewable generation, electricity price, connected-fleet availability, and aggregate battery SOC.

The current implementation uses a safe rule-based controller as the baseline. It is designed so that optimization and ANFIS control can be added later without removing the physical SOC, charger-power, and transformer constraints. Short-duration G2V and V2G converter outputs are presented separately as reference behavior for the planned detailed charger validation layer.

### Project objectives

- Reduce charging contribution during feeder peaks and expensive tariff periods.
- Increase the use of locally available solar and wind generation.
- Maintain battery SOC within safe minimum and maximum limits.
- Provide controlled V2G support without sacrificing departure readiness.
- Create a clear path from a baseline controller to optimization and ANFIS.

### Key baseline indicators

|                       |            |                       |         |
|-----------------------|------------|-----------------------|---------|
| Peak grid demand      | 446.2 kW   | Peak-to-average ratio | 2.101   |
| Grid energy imported  | 5151.0 kWh | Grid energy exported  | 0.0 kWh |
| Renewable utilization | 100.0%     | Final fleet SOC       | 81.2%   |

These values describe the present rule-based baseline and are not final optimized results.

## 02 System architecture



### Supervisory simulation

The supervisory layer uses a 24-hour horizon because grid demand, renewable generation, tariffs, EV arrival patterns, and departure SOC targets change slowly. A 15-minute step is appropriate for scheduling and keeps scenario studies computationally efficient.

### Detailed converter validation

A switching converter requires microsecond-level electrical simulation. It should therefore test selected charging, idle, and discharging commands over approximately 0.1 to 1 second rather than simulate the entire day. The two layers exchange signed power or current references.

### Power direction convention

| Command      | Operating state | Energy direction              |
|--------------|-----------------|-------------------------------|
| $P_{EV} > 0$ | G2V charging    | Grid to EV battery            |
| $P_{EV} = 0$ | Idle            | No requested battery exchange |
| $P_{EV} < 0$ | V2G discharging | EV battery to grid            |

### Grid power balance

$$P_{grid} = P_{base} - P_{renewable} + P_{EV}$$

This sign convention makes charging an additional grid demand and V2G a reduction in net demand.

## 03 Model inputs, constraints, and control

### Input signals

| Signal             | Purpose                            | Unit    |
|--------------------|------------------------------------|---------|
| Base load          | Residential feeder demand          | kW      |
| Renewable power    | Combined solar and wind production | kW      |
| Electricity price  | Time-of-use charging incentive     | EUR/kWh |
| Connected fraction | Available part of the EV fleet     | 0 to 1  |
| Fleet SOC          | Aggregate stored-energy state      | 0 to 1  |

### Baseline fleet parameters

| Parameter            | Value  | Parameter            | Value    |
|----------------------|--------|----------------------|----------|
| Fleet size           | 50 EVs | Total capacity       | 3000 kWh |
| Maximum charge power | 360 kW | Maximum V2G power    | 250 kW   |
| Initial SOC          | 45%    | Target SOC           | 80%      |
| Minimum SOC          | 20%    | Maximum SOC          | 95%      |
| Charge efficiency    | 94%    | Discharge efficiency | 92%      |
| Transformer limit    | 500 kW | Sample time          | 0.25 h   |

### Decision priority

- Protect the battery at minimum and maximum SOC.
- Absorb otherwise surplus renewable power.
- Charge during low-price periods when SOC is below target.
- Discharge during feeder peaks or high-price periods when sufficient energy is available.
- Apply charger and transformer limits after the scheduling decision.

## 04 Twenty-four-hour smart-grid results

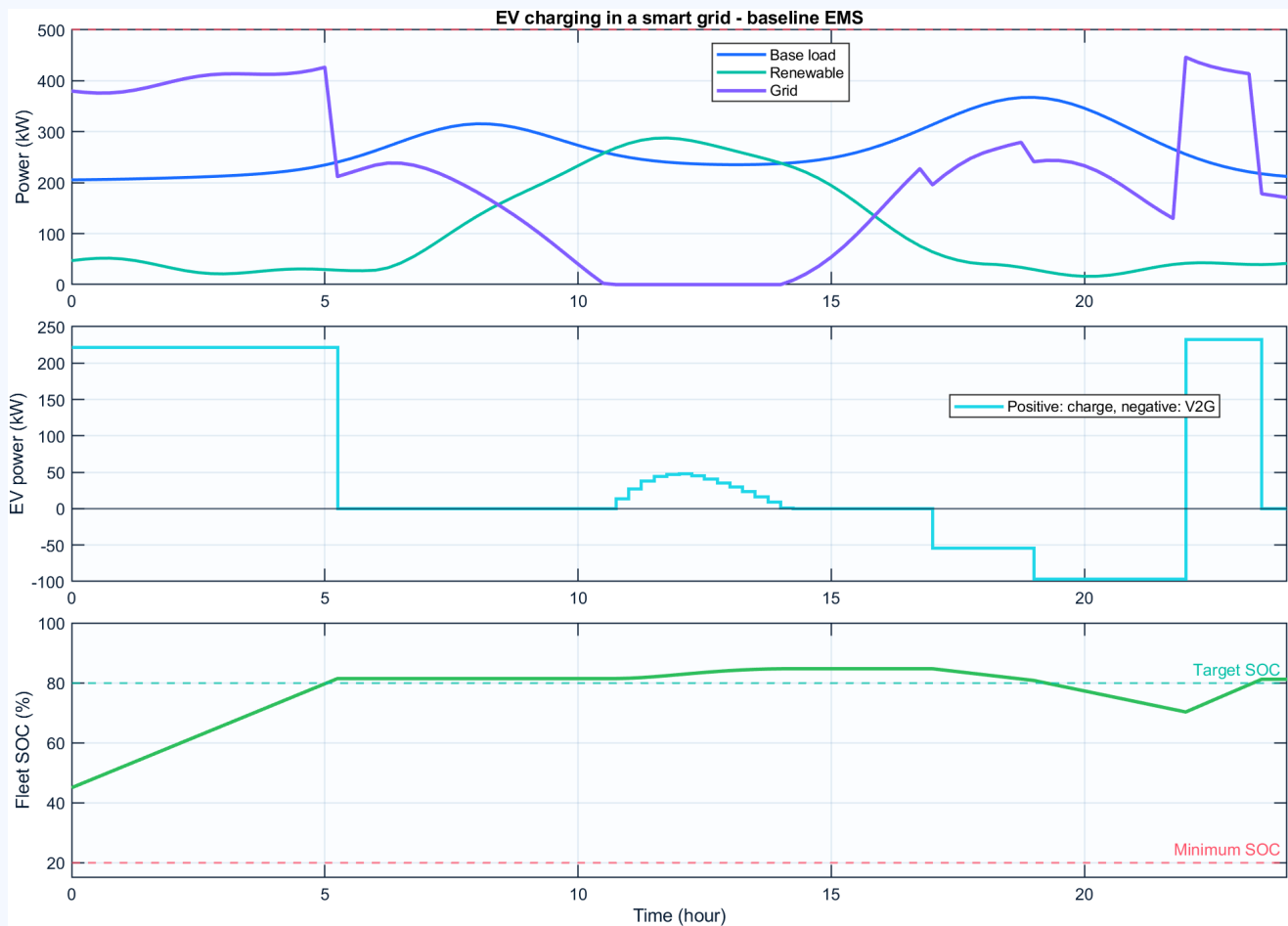


Figure 1. Rule-based baseline: feeder demand, renewable generation, net grid power, EV charging/V2G command, and aggregate fleet SOC.

### Interpretation

The fleet charges strongly during periods with high connection and available capacity. Around the evening peak, the command becomes negative and provides V2G support. SOC remains above the minimum limit and returns close to the 80% target by the end of the day. The grid demand remains below the 500 kW transformer limit in the baseline case.

This controller is a safe reference case. A fair final evaluation should compare it against uncontrolled charging, no-EV operation, constrained optimization, and ANFIS using identical input scenarios.

## 05

## Detailed reference output: G2V charging

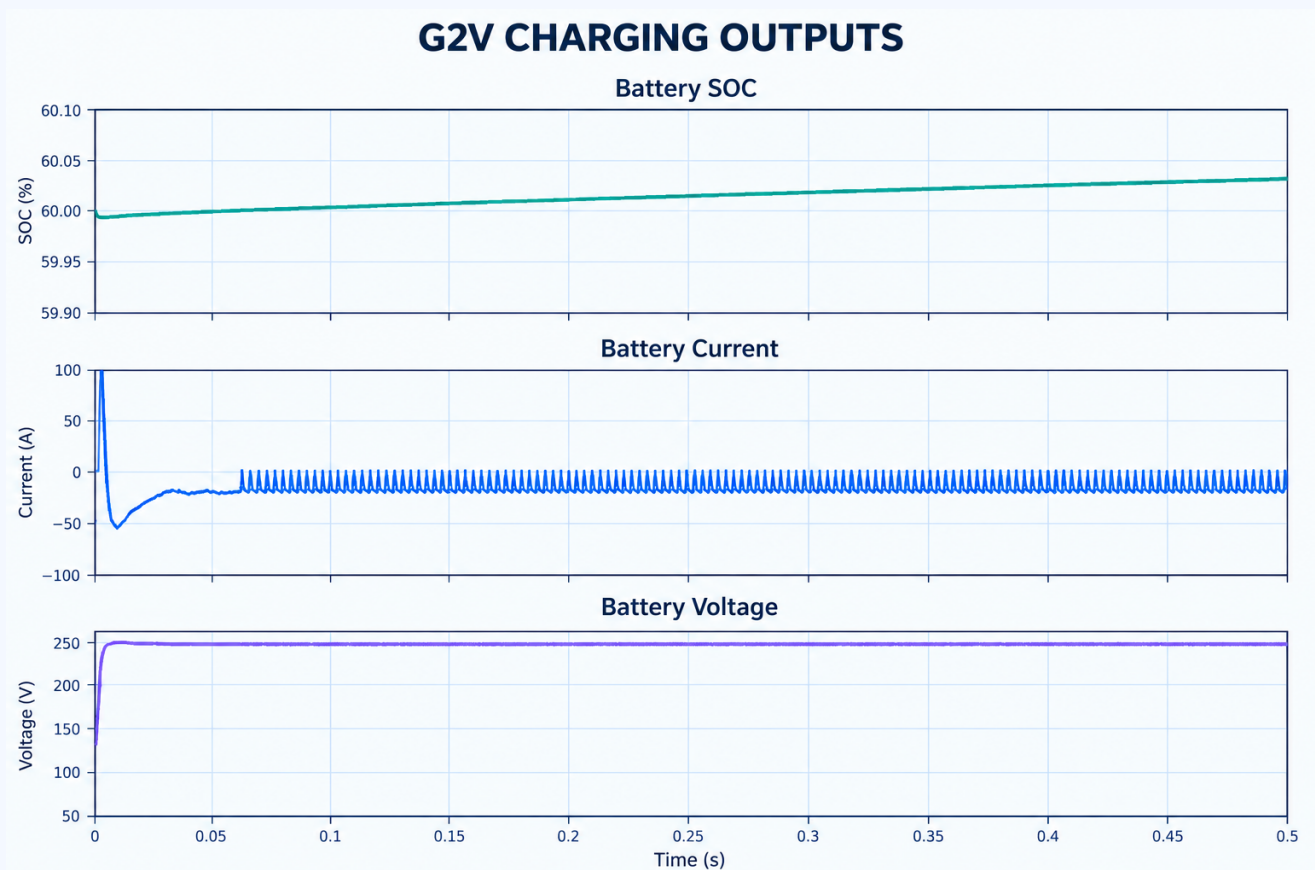


Figure 2. Restyled reference output for grid-to-vehicle operation. The three traces retain the SOC, current, and voltage behavior of the supplied converter simulation output.

### Observed behavior

- Battery SOC rises gradually, confirming net energy transfer into the battery.
- Battery current shows an initial transient followed by controlled switching ripple.
- Battery voltage rapidly approaches its regulated operating value.

The short electrical time scale is suitable for converter stability and waveform inspection, but it is not intended to represent a complete charging session.

## 06

## Detailed reference output: V2G discharging

## V2G DISCHARGING OUTPUTS

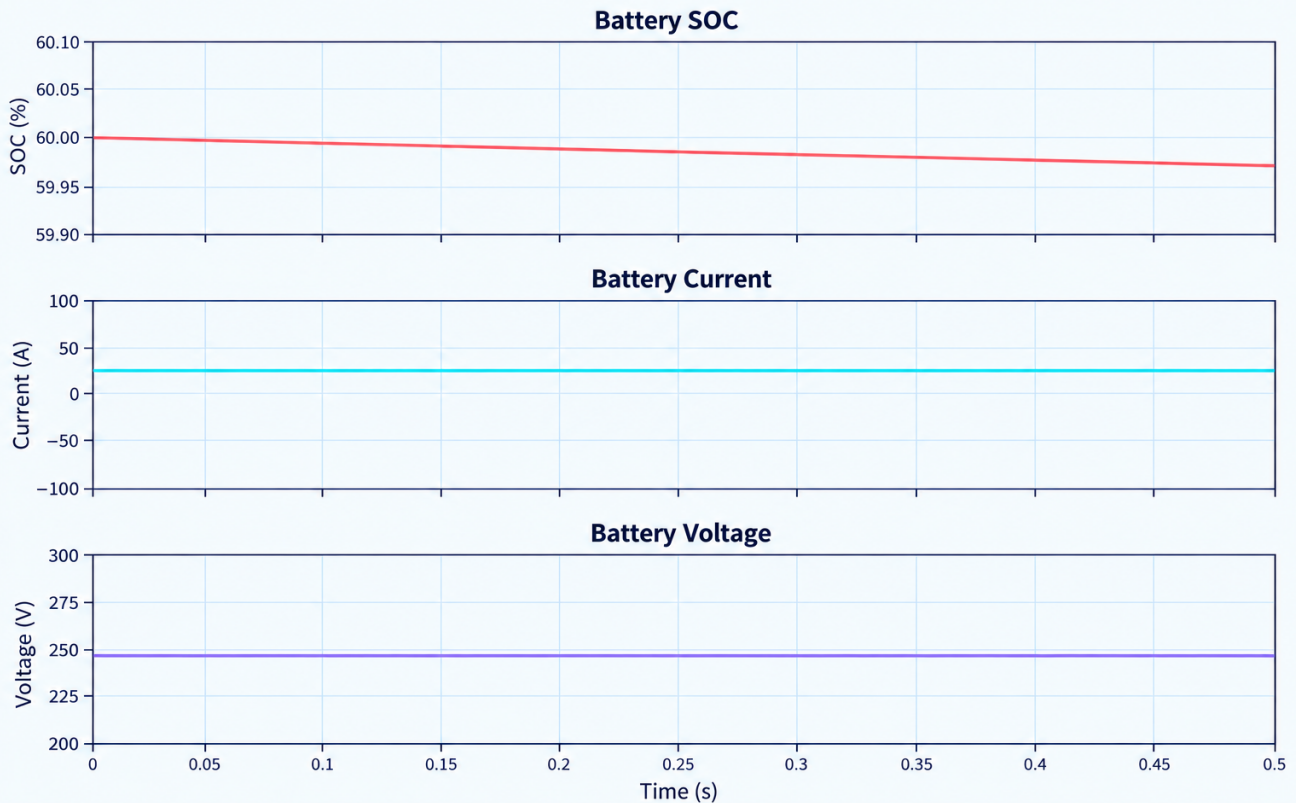


Figure 3. Restyled reference output for vehicle-to-grid operation. The SOC decreases slowly while current and voltage remain approximately constant over the displayed interval.

## Observed behavior

- The downward SOC trend is consistent with energy export from the EV battery.
- Approximately constant battery current indicates controlled discharge operation.
- Stable battery voltage supports controlled bidirectional power transfer.

These detailed traces are reference outputs restyled from the supplied poster. They should be replaced by freshly exported results after the detailed charger is connected to the supervisory command interface.

## 07 Integration and intelligent-control plan

### Recommended interface

The day-ahead layer should export selected signed power references. A command mapper converts each reference into charger mode, enable, and current magnitude. The detailed charger then returns measured power, DC-link voltage, battery current, efficiency, THD, and battery stress.

| Stage | Implementation task                        | Output                       |
|-------|--------------------------------------------|------------------------------|
| 1     | Validate baseline smart-grid model         | Reference metrics            |
| 2     | Add uncontrolled and no-EV cases           | Fair comparison              |
| 3     | Solve constrained day-ahead schedules      | Optimal EV commands          |
| 4     | Train and validate Sugeno ANFIS            | Fast real-time policy        |
| 5     | Add three-state detailed charger interface | Charge / idle / V2G          |
| 6     | Run selected switching test cases          | Efficiency and power quality |
| 7     | Return measured maps to scheduler          | Improved system accuracy     |

### ANFIS definition

Recommended inputs: base load, renewable power, electricity price, fleet SOC, and connected fraction.  
Recommended output: aggregate signed EV power in kW. The learned output must always pass through an independent safety supervisor before reaching the plant.

### Validation scenarios

- High solar production with low fleet SOC.
- Evening feeder peak with sufficient V2G reserve.
- Minimum-SOC and maximum-SOC boundary protection.
- Sudden charge-to-idle-to-discharge command changes.
- Forecast errors and changes in vehicle availability.

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# Conclusions and reproducibility

The project now provides a reproducible 24-hour smart-grid baseline and a clear framework for converter-level V2G validation. The aggregate model captures the operational decisions that matter across a day, while the detailed short-duration outputs show the electrical behavior expected during charging and discharging.

The next major research contribution should be a constrained optimizer and an ANFIS controller trained from optimal schedules. Final claims should be based on repeatable comparisons against uncontrolled charging and the current rule-based baseline.

## How to reproduce the current result

|   |                                                                                          |
|---|------------------------------------------------------------------------------------------|
| 1 | Open MATLAB R2024a or R2024b.                                                            |
| 2 | Set Current Folder to the repository root.                                               |
| 3 | Run: <code>run("scripts/run_ev_smart_grid.m")</code>                                     |
| 4 | Review <code>results/baseline_results.png</code> and <code>baseline_results.mat</code> . |

## Files supplied

| Path                                             | Purpose                                                 |
|--------------------------------------------------|---------------------------------------------------------|
| <code>models/supervisory/EV_SmartGrid.slx</code> | Supervisory Simulink model                              |
| <code>scripts/*.m</code>                         | Initialization, control, SOC, model builder, and runner |
| <code>results/</code>                            | Reproducible baseline result and numerical MAT-file     |
| <code>images/</code>                             | Redesigned G2V and V2G output figures                   |
| <code>report/</code>                             | This project report                                     |

Simulation and documentation package prepared for GitHub presentation. The model is intended for education and research, not direct hardware control.